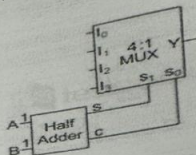


a. $Y = I_0$
(b.) $Y = I_1$
 c. $Y = I_2$
 d. $Y = I_3$



[15 Marks]

Simplify by using De Morgan's Law

$$= (A'B'C + \cancel{A'B'BD}) + A'B'C$$

$$= (AB'C + A'B'C)$$

~~$$= [B(A \cup A)]C$$~~

$$= AB'CC + A'B'C$$

$$= A B' C + A B' C$$

$$= B'c(A+A') = 1$$

$= BC$

Question 3...

Question 3.

A- Simplify the function F using Boolean laws:

$$F = (x + y + z)(x + y' + z)(x + y + z')$$

$$= f(x) = (x^2 + y^2 + z^2 + y^2 + z^2)(x + y + z)$$

$$(x + ny + nz + by + z)(n + y + z')$$

$$[n(\cancel{1+y+z} + 1+y+z) + z](n+y+z)$$

$$n(1) + z(n + y + z')$$

$$n + z(n + y + z')$$

Page 3 of 8

$$= \frac{nx + ny + nz + x^2 + y^2 + z^2}{x^2 + y^2 + z^2}$$

$$x + xy + xz + 7y$$

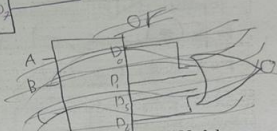
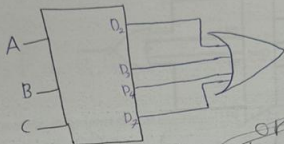
3

c. Implement the following function using the appropriate Decoder [2 Marks]

$$F' = \sum (0, 1, 5, 6)$$

$$F = \sum (2, 3, 4, 7)$$

2



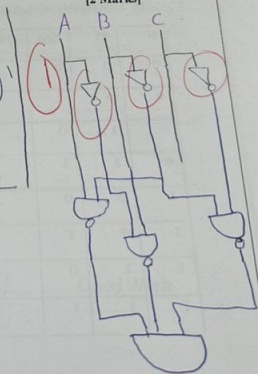
Question 5 [4 Marks]

A- Design the following Boolean Expression (F) by using NAND gates only

$$F = B'C' + AB' + A'B$$

[2 Marks]

$$\begin{aligned} F &= (B'C')' + \overline{AB} + (A'B)' \\ &= ((B'C')' + (AB)') + (A'B)' \\ &= (B'C')(AB)' + (A'B)' \\ &= (B'C')(AB)'(A'B)' \\ &= (B'C')'(AB)'(A'B)' \end{aligned}$$



2. Simplify the following function using 2-Map in a Minimum form. (Sum of Product SOP)

$$F(X, Y, Z, W) = WXY' + WY' + WYZ' \quad [5 \text{ Marks}]$$

Handwritten Karnaugh Map for $F(X, Y, Z, W)$:

	00	01	11	10
00				1
01				1
11			1	1
10	1	1	1	1

Groupings shown: A vertical group of four 1s in the last column (W=1), a horizontal group of four 1s in the last row (Y=0), and a group of two 1s in the bottom row, middle columns (X=1, Z=1).

$$F = W\bar{X} + WY' + YZ'$$

Question 3 [5 Marks]

Implement the following Boolean function $F(A, B, C, D) = \sum(0, 1, 3, 5, 9, 10, 11, 15)$

using 4:1 MUX. With A, B as Select lines

- a. Find F using the truth table

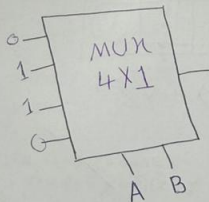
[1 Marks]

(1)

A	B	C	D	F	
0	0	0	0	1	0
0	0	0	1	1	1
0	0	1	0	0	2
0	0	1	1	1	3
0	1	0	0	0	4
0	1	0	1	1	5
0	1	1	0	0	6
0	1	1	1	0	7
1	0	0	0	0	8
1	0	0	1	1	9
1	0	1	0	1	10
1	0	1	1	1	11
1	1	0	0	0	12
1	1	0	1	0	13
1	1	1	0	0	14
1	1	1	1	1	15

b. Design of the Multiplexer (4:1 MUX.)

[2 Marks]



ED	ED	CD	CD
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

d	d	D
0	1	0
2	3	0
4	5	0
6	7	0
8	9	0
10	11	0
12	13	0
14	15	0

$$4\frac{1}{2}$$

Section 1: True/False & Multiple-choice question
Question 1:A-Mark the correct Answer by True or False

(1/2 mark for each)

(1/2 mark for each)

(1/2 mark for each)

1. What is a combinational circuit?

- a) A circuit whose output depends only on the current input
- b) A circuit that has memory elements
- c) A circuit that performs sequential operations
- d) A circuit that uses feedback to produce outputs

2. How many 4x16 Decoder using 2x4 Decoders

- a) 4
b) 1
c) 8
d) 5

3. Construct a 32×1 multiplexer. How many 2×1 multiplexers are required to construct?

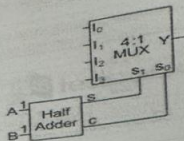
- a) 8
b) 15
c) 16
d) 31

4. The difference between Half Adder and Full Adder is:

- a) half adder has two inputs while full adder has four inputs
 b) half adder has two inputs while full adder has three inputs
 c) half adder has one inputs while full adder has two inputs
 d) None of the above

4

- X



[15 Marks]

[2 Marks]

$$F = (A, B'(C + B, D) + A', B'), C$$

2

~~$$= \overline{B(A+C)}$$~~

$$= AB'c + A'B'c$$

$$= B'c(A+A') \rightarrow 1$$

$$= B'C$$

Question 3.

[4 Marks]

A- Simplify the function F using Boolean laws:

[2 Marks]

$$F = (x + y + z)(x + y' + z)(x + y + z')$$

$$= f(x) = (x^2 + y^2 + z^2 + x^2 y^2 + y^2 z^2 + z^2 x^2)(x^2 + y^2 + z^2)$$

$$[n(\cancel{1} + y + x + z) + z](n + y + z)$$

$$n(1) + z(n + 1 + z')$$

$$-n + z(n + y + z')$$

Page 3 of 8

$$= nx + ny + nz + x^2 + y^2 + z^2$$

$$x + xy + xz^4 + xz + 7y$$

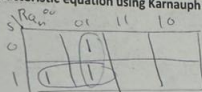
$$n(1 + \underbrace{y + z}_{\text{odd}} + \underbrace{z}_{\text{even}}) + zy$$

$$\{x + 2y\}$$

- 2) Find the characteristic equation using Karnaugh Map

[2Marks]

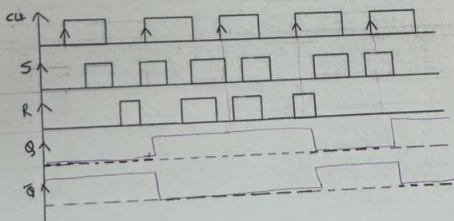
- K-Map



$$Q(n+1) = S\bar{R} + \bar{R}Q_n$$

- 3) Complete the state diagram of the SR Flip Flop

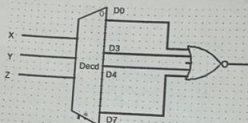
[2Marks]



$$Q(n+1) = \bar{S}\bar{R}Q_n + \bar{S}R\bar{Q}_n + S\bar{R}Q_n$$

Good Work

B- Using the 3 to 8 Decoder below to illustrate F and F' (complement) in sort of Minterm(SOP) or Maxterm(POS) [2 Marks]



$$F = \sum (3, 4, 7) = \prod (1, 2, 5, 6)$$

$$F' = \sum (0, 3, 4, 7)$$

Section 3: Sequential Circuit

[5 Marks]

Given the following positive Edge Triggered SR Flip-flop.

1) Find Q(n+1) using the truth table [1 Marks]

S	R	Qn	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

